Citation Navigator: Streamlining Literature Review Through Interactive Citation Analysis

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Abstract

Reading and evaluating research papers is a time-intensive task, made more difficult by the need to assess the relevance of dozens or even hundreds of cited works. We present Citation Navigator, a web application that streamlines citation analysis by combining automated PDF parsing, metadata enrichment, and AI-generated summaries into a unified interface. Users upload a research paper and are presented with an interactive PDF viewer in which citation markers are highlighted. Clicking a citation opens a panel displaying a concise summary of the cited work, its metadata, and the context in which it is cited in the source paper. Citation Navigator uses GROBID for structured PDF extraction, OpenAlex for metadata and abstract retrieval, and the Groq or Anthropic Claude API to generate contextual summaries grounded in both the abstract of the cited work and its usage context. We discuss key implementation challenges including incomplete citation marker extraction and limited abstract availability, and describe the solutions developed to address them. Evaluation on a set of research papers demonstrates reliable citation extraction and metadata coverage, enabling researchers to quickly assess the relevance of cited works without leaving the reading environment.

Keywords

Citation Analysis, Research Paper Navigation, PDF Extraction, Large Language Models, Information Retrieval

1 Introduction

Research papers are the primary medium through which scientific knowledge is communicated and built upon. A single paper can cite dozens or even hundreds of prior works, each representing a potentially relevant thread for a reader to follow. For researchers, students, and practitioners attempting to survey a field or evaluate a paper's contributions, this presents a significant challenge: assessing which cited works are worth reading in full requires either prior familiarity with the literature or the time-consuming process of locating and skimming each cited paper individually. In practice, most readers simply skip this step, potentially missing important context or misunderstanding the significance of a contribution.

To help combat this problem, we present Citation Navigator, a web application that addresses this gap by bringing citation analysis directly into the reading experience. Users upload a PDF of a research paper and are presented with an interactive side panel containing the cited paper's metadata, a concise AI-generated summary, and the specific sentence or passage in which the citation appears in the source paper. This allows readers to quickly assess the relevance and contribution of a cited work without interrupting their reading flow.

Citation Navigator is built on a pipeline combining GROBID for structured PDF parsing, OpenAlex for metadata and abstract retrieval, and the Anthropic Claude API for generating contextual summaries. In building the system we encountered and addressed several practical challenges, including incomplete citation marker extraction, limited abstract availability across metadata sources, and the extraction of citation context from raw PDF text. The remainder of this paper describes the system design, implementation details, evaluation, and usage of Citation Navigator.

2 Motivation

2.1 Key Issues

The volume of academic literature has grown dramatically in recent decades. As of 2023, an estimated 4 million peer-reviewed articles are published annually, and this number continues to rise. For researchers, keeping up with the literature in even a narrow subfield requires reading and evaluating a continuous stream of new work. A single survey paper may cite 200 or more prior works, and even a focused conference paper typically cites 30–50 references. To efficiently review a paper, readers encountering an unfamiliar citation mid-paragraph must make a quick judgment: is this citation foundational to understanding the current work, or is it peripheral? Is the cited result being used as a direct building block, or merely acknowledged as prior art? Answering these questions currently requires the reader to stop, locate the cited paper, and skim it, which interrupts the reading flow and compounds across dozens of citations.

This problem is particularly acute for researchers entering a new area, graduate students conducting literature reviews, and practitioners evaluating whether an academic result applies to their work. In all of these cases, the ability to quickly assess the relevance and contribution of a cited work without leaving the current paper would meaningfully reduce the time and cognitive overhead of reading research.

2.2 Existing Work

Existing tools offer partial solutions to this problem. Reference managers such as Zotero and Mendeley help organize collections of papers but offer little assistance in understanding the relationships between them. Citation graph tools such as Connected Papers visualize how papers relate to one another at a high level but do not provide in-context analysis of how a specific citation is used within a paper. Semantic Scholar provides paper summaries and citation counts but requires leaving the reading environment entirely.

Citation Navigator aims to bridge these gaps by offering an interactive reader that provides contextual analysis, bringing key information as close as possible to the reader.

3 System Architecture and Implementation

3.1 Overview

Citation Navigator follows a client-server architecture in which a React frontend communicates with a Python Flask backend over a REST API. The backend orchestrates a multi-stage pipeline: structured PDF parsing via GROBID, citation extraction and normalization, metadata and abstract retrieval via OpenAlex, and on-demand AI-generated summarization of the user's choice.

3.2 Frontend

The frontend is implemented in React and provides three primary interfaces: a PDF upload screen, a PDF viewer, and a citation detail panel. Upon upload, the PDF is transmitted to the Flask backend and the viewer renders the document. Once the backend extracts the citations, a clickable list of extracted citations is displayed in a side panel. When a citation in the side panel is clicked, the citation's metadata, the context sentence in which it appears, and an AI-generated summary are displayed. The summary is fetched lazily on first click and cached client-side for subsequent accesses, avoiding unnecessary API calls.

3.3 Backend

The backend is implemented in Python using the Flask framework and exposes two primary endpoints: one for PDF upload and processing, and one for pulling single citation data. On receiving an uploaded PDF, the backend initiates the extraction pipeline and returns the structured citation data to the frontend. Summarization requests are handled separately, allowing the frontend to render the PDF viewer immediately while summaries are generated on demand.

3.4 PDF Parsing and Citation Extraction

PDF parsing is handled by GROBID, an open-source machine learning library for extracting and structuring information from scholarly documents. Citation Navigator deploys GROBID using the CRF-only Docker image, which foregoes deep learning models in favor of conditional random field models that offer faster inference with minimal accuracy tradeoff for citation extraction tasks. The backend submits uploaded PDFs to GROBID's `processFulltextDocument` endpoint, which returns a TEI XML document containing the structured full text, a parsed bibliography, and bounding box coordinates for each citation marker in the document.

The TEI XML is parsed to extract two artifacts: a reference list mapping each citation ID to its structured metadata fields, and a marker map associating each in-text citation marker (e.g. `[1]`, `[3-5]`) to its corresponding reference list entries. In practice, GROBID often fails to extract marker strings for some references. To address this, Citation Navigator infers missing markers from the sequential ordering of GROBID's internal reference IDs, exploiting the fact that GROBID assigns IDs `b0`, `b1`, `b2`, ... in bibliography order, which corresponds directly to numeric citation markers `[1]`, `[2]`, `[3]`, ...

In addition to markers extracted from GROBID's TEI output, Citation Navigator performs a secondary pass over the raw PDF text to recover any markers that GROBID failed to detect. A regular expression scans the full text for patterns matching numeric citation markers, including standalone markers (`[1]`), comma-separated lists (`[1, 3, 5]`), and hyphenated ranges (`[1-3]`). Any markers discovered in this pass that are absent from GROBID's output are merged into the marker map, with their reference list entries inferred from the sequential ID convention described above. This two-pass approach improves marker coverage and ensures that citations missed by GROBID's parser are still surfaced in the viewer.

3.5 Metadata Extraction and Enrichment

For each extracted citation, Citation Navigator retrieves metadata and abstracts exclusively from the OpenAlex API. OpenAlex is an open catalog of scholarly literature providing broad coverage of academic papers across disciplines, including title, authors, year, venue, and citation count. A key advantage of OpenAlex over alternatives such as Semantic Scholar is its consistent abstract availability. OpenAlex encodes abstracts as inverted indexes, a format adopted for copyright compliance in which each word is mapped to its positions in the abstract rather than stored as raw text. Citation Navigator reconstructs the abstract from this inverted index at query time before passing it to the summarization pipeline. For papers not found in OpenAlex, the summarization prompt falls back to the citation context alone, relying on the LLM to infer the contribution of the cited work from how it is referenced in the source paper.

3.6 LLM-generated Contextual Summaries

Summaries are generated on demand when a user clicks a citation marker. The backend constructs a prompt containing the cited paper's title, authors, abstract (if available), and the citation context extracted from the source paper. This prompt is submitted to the active LLM backend, which returns a structured response including a concise summary of the cited work, an explanation of how it is used in the context of the source paper, and an assessment of its relevance to the source paper's contribution.

Citation Navigator supports two LLM backends: the Anthropic Claude API and Groq. Both backends receive the same structured prompt and return summaries in the same format, and the active backend is configurable at runtime via an environment variable. Groq is included as the default option for users who do not have access to an Anthropic API key, as Groq provides a generous free tier suitable for typical usage volumes. The Anthropic Claude API is recommended for higher quality summaries. The abstraction over both backends is kept deliberately thin, making it straightforward to add additional LLM providers in the future. Summaries are generated lazily, so the backend does not pre-generate summaries for all citations on upload. This design choice avoids a large upfront latency and API cost on upload, since users typically interact with only a subset of citations in any given session. Generated summaries are cached server-side by citation ID so that repeated clicks on the same citation within a session do not trigger redundant API calls.

4 Evaluation

Given the generative nature of Citation Navigator's primary output, formal quantitative evaluation of summary quality is difficult to conduct at scale. Instead, we evaluate the system along two measurable dimensions: citation extraction accuracy and runtime performance.

4.1 Citation Extraction Accuracy

Citation extraction accuracy was assessed by manually comparing the citations extracted by Citation Navigator against the ground truth reference lists of a sample of research papers. Across the papers tested, Citation Navigator achieved an extraction accuracy of approximately 90%, with missed citations attributable primarily to unusual bibliography formatting and two-column layout artifacts that confuse GROBID's parser. The two-pass extraction approach was found to meaningfully improve coverage over GROBID alone, recovering markers that the structured parser failed to detect.

4.2 Runtime Performance

End-to-end processing time from PDF upload to an interactive viewer with citation highlights was measured across a sample of papers of varying lengths. The lazy loading architecture contributes significantly to perceived responsiveness. Individual citation summary generation, including the OpenAlex metadata lookup and LLM inference, averages approximately 1.2 seconds per citation. This latency is non-negligible but also not unreasonable for a one time bulk load: loading time for a very large survey paper with 200 citations took approximately 4 minutes, which could be improved with better resources.

5 Demo

Citation Navigator is designed to be intuitive to use, requiring no configuration or prior knowledge of the underlying pipeline. This section walks through a typical usage session illustrated with screenshots of the running system.

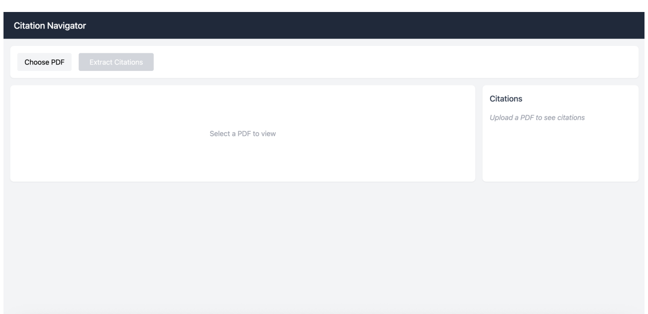

Figure 1: The Citation Navigator landing screen, where users upload a research paper PDF.

A user begins by navigating to the Citation Navigator web interface, shown in Figure 1. The landing screen presents a clean upload interface where the user selects a PDF of a research paper from their local machine.

Upon upload, the system displays a processing screen (Figure 2) while the backend pipeline runs: GROBID parses the PDF, citation markers are extracted and normalized, and metadata is retrieved from OpenAlex for each identified reference.

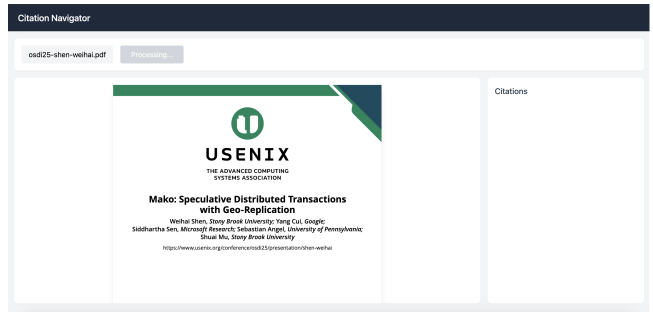

Figure 2: The processing screen displayed while GROBID parses the PDF and citation metadata is retrieved from OpenAlex.

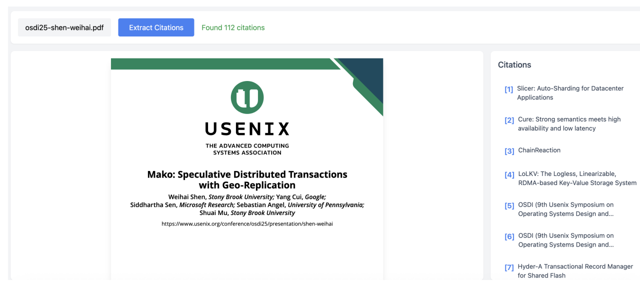

Figure 3: The interactive PDF viewer with extracted citations shown in the sidebar. Clicking any citation opens the citation detail panel.

Once processing is complete, the PDF is rendered in the interactive viewer with all extracted citations displayed in the sidebar, as shown in Figure 3. The user can scroll through the paper naturally and click any citation to show details.

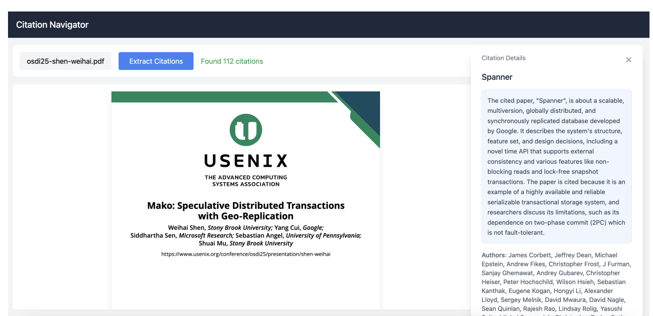

Figure 4: The upper half of the citation detail panel, showing the AI-generated summary and cited paper metadata.

Clicking a citation opens the detail panel, shown in Figures 4 and 5. The upper half of the panel displays an AI-generated summary of the cited work along with metadata retrieved from OpenAlex, including the title, authors, publication year, and venue. The lower half displays the abstract of the cited paper and the context extracted from the source paper, giving the reader an immediate

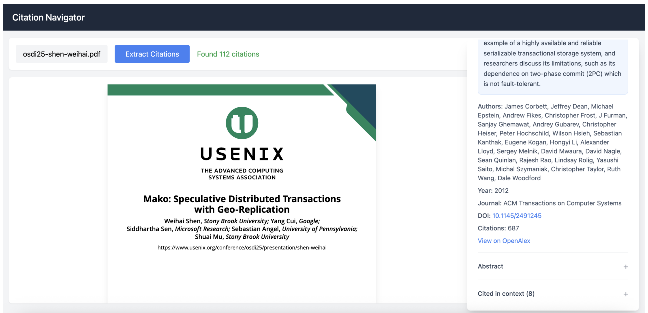

Figure 5: The lower half of the citation detail panel, showing the abstract of the cited work and the context sentence in which the citation appears in the source paper.

sense of how and why the work is cited. Summaries are generated lazily on first click and cached for the duration of the session, keeping query latency negligible for repeated accesses.

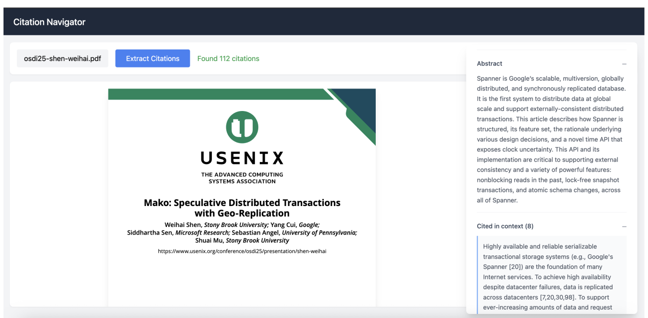

Figure 6: The citation detail panel with the abstract section expanded. Both the abstract and citation context are collapsible, allowing users to focus on the information most relevant to them.

Both the abstract and citation context fields in the detail panel are collapsible, as shown in Figure 6. This allows users to expand only the sections they find relevant, keeping the panel uncluttered when a quick summary is sufficient and enabling a deeper dive when the full abstract or surrounding context is needed.

6 Conclusion

Citation Navigator demonstrates that a functional and useful citation analysis tool can be built within a limited development timeframe by composing existing open-source and API-based components into a coherent pipeline. The system successfully extracts citations from uploaded research papers, enriches them with metadata and abstracts from OpenAlex, and delivers AI-generated contextual summaries on demand all within an interactive reading interface that keeps the user's focus on the paper at hand.

Several directions for future development remain. The current deployment runs GROBID's CRF-only models locally due to hardware constraints, and transitioning to cloud or GPU-backed hosting would enable the use of GROBID's full deep learning models, which offer meaningfully higher citation extraction accuracy and reduced processing latency. Additionally, like all LLM-powered systems, Citation Navigator is susceptible to hallucinated content in generated summaries. Investigating and implementing mitigation strategies such as grounding checks against retrieved abstracts or confidence-based flagging would improve the reliability of the summaries and increase user trust in the system.

A Project Code

The code for this project, including detailed installation instructions, is available at: https://github.com/brad-zeng/citation-navigator